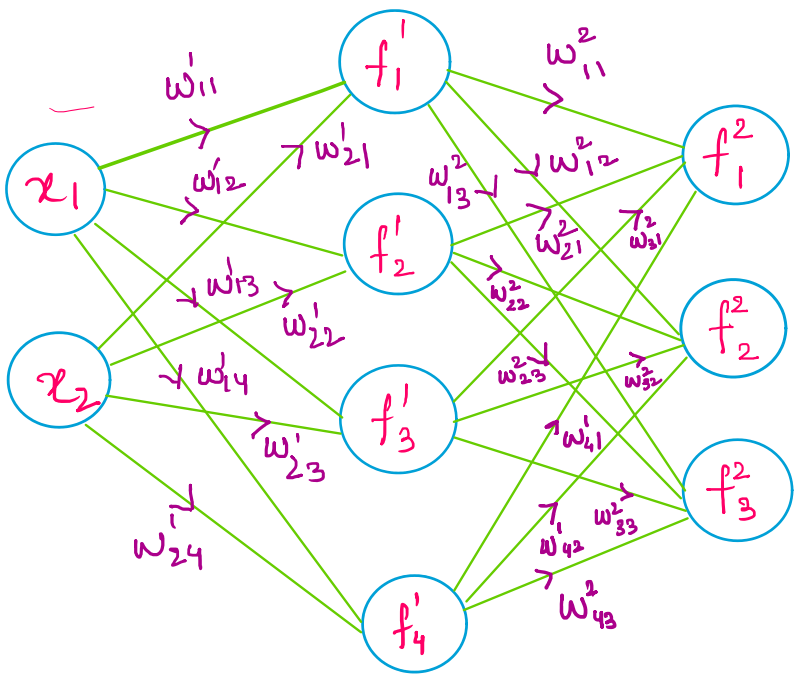
Neural Networks-3

★ Multi Layer Perceptron (MLP):-

NN formed by adding a layer of neurons in b/w an i/p and o/p layer. This intermediate layer is known as a hidden layer.
As the computation of these layers is not accessible from outside, hence these layers are called hidden layers.



weight matrix :

Layer-1

$$\begin{bmatrix} w_{11}^1 & w_{12}^1 & w_{13}^1 & w_{14}^1 \\ w_{21}^1 & w_{22}^1 & w_{23}^1 & w_{24}^1 \end{bmatrix}_{2\times 4}$$

Layer-2

$$\begin{bmatrix} w_{11}^2 & w_{12}^2 & w_{13}^2 \\ w_{21}^2 & w_{22}^2 & w_{23}^2 \\ w_{31}^2 & w_{32}^2 & w_{33}^2 \\ w_{41}^2 & w_{42}^2 & w_{43}^2 \end{bmatrix}_{4\times 3}$$

i/p layer | Hidden Layer (Layer-1) | O/P Layer (Layer-2)

i/p layer is not considered as a computation layer
Hidden and O/P layers are considered → neurons
computation ← activation function

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* Activation Functions in case of Hidden Layers :-

① **Hyperbolic Tanget** :- It is a shifted version of sigmoid.

It works better than sigmoid.

I/P range → $-\infty$ to ∞

O/P range → -1 to $+1$

Formula :

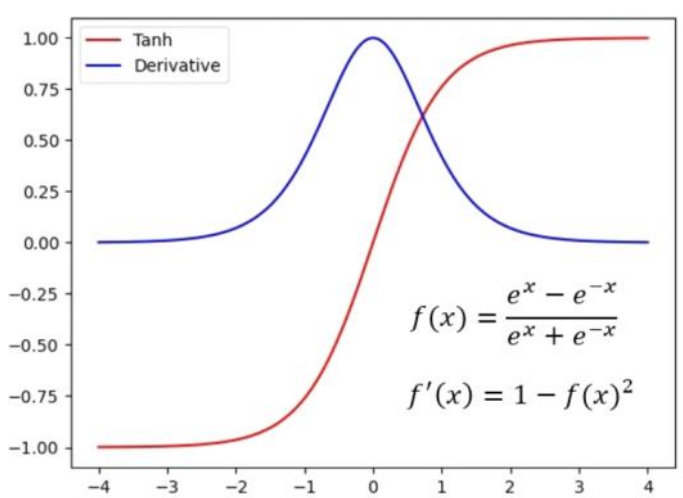
$$\tanh(z) = \frac{e^{z} - e^{-z}}{e^{z} + e^{-z}}$$

& the derivative –

$$\tanh'(z) = 1 - \tanh^2(z)$$

The derivative lies b/w 0 and 1.

This function is used when we want O/P to lie in the range -1 to $+1$



Issue with sigmoid and tan h:-

Vanishing Gradients – Problem that occurs when gradients become too small [as the derivative lies b/w 0 and 1]

↓

Product will be very small

↓

weight update very slow → slow learning

* Ideal Activation :-

1. Differentiable
2. Non-Linear
3. Easy Calculation
4. Gradient ≥ 1 for big range of z

⇓

2. Rectified Linear Unit [ReLU] :-

An activation function for which –

$$ReLU = \begin{cases} z, & \text{if } z > 0 \\ 0, & \text{if } z \leq 0 \end{cases}$$

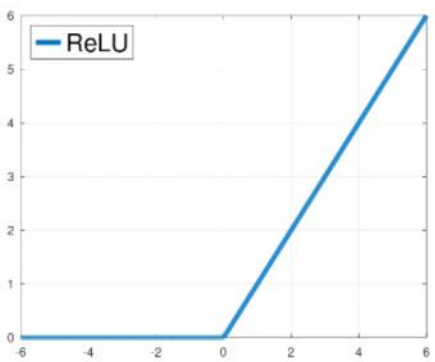
& the derivative –

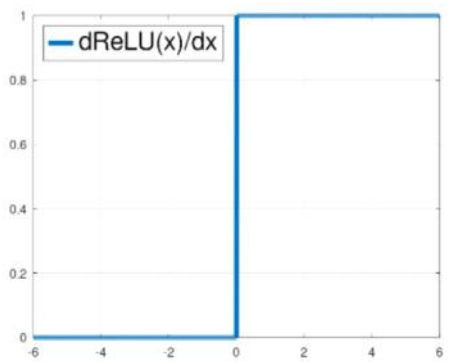
$$ReLU'(z) = \begin{cases} 1, & \text{if } z > 0 \\ 0, & \text{if } z < 0 \end{cases}$$

∵ it is non-differentiable at $z=0$ hence taking an approximation

$$\text{ReLU}'(z) = \begin{cases} 1, & \text{if } z>0 \\ 0, & \text{if } z \leq 0 \end{cases}$$ → approximation

Therefore, ReLU activation function is very fast to compute and is widely used in NN.





Though ReLU activation is widely used, but it comes with a problem.

Issue with ReLU :-

If anyone 1 term of the derivative while calculating the loss becomes 0, the entire term will become 0. This is known as dying ReLU.

↳ Dying of Neurons

Solution to this problem → Leaky ReLU

③ Leaky ReLU :-

This activation is similar to ReLU but comes with a

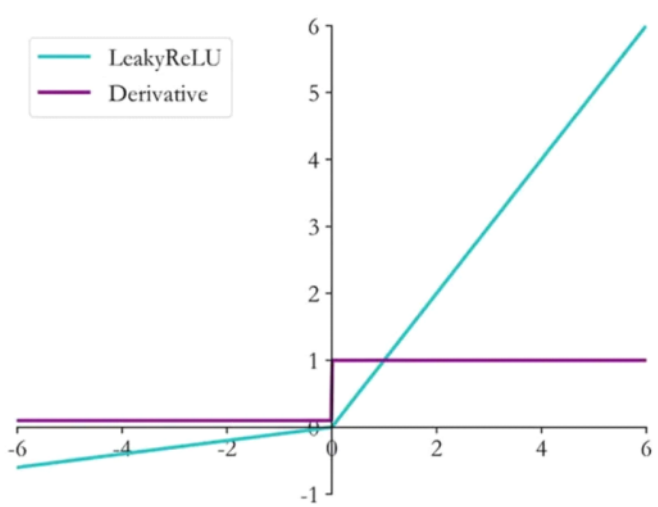
Small modification.
For $z<0$, a small gradient (α) is added.

$$\text{Leaky ReLU} = \begin{cases} z, & \text{if } z>0 \\ \alpha z, & \text{if } z\leq 0 \end{cases}$$

& its derivative –

$$\text{Leaky ReLU}'(z) = \begin{cases} 1, & \text{if } z>0 \\ \alpha, & \text{if } z\leq 0 \end{cases}$$

This solves the problem of dying ReLU. This variation of ReLU has a small +ve slope in the -ve area.
It enables backpropagation even for -ve i/p values
It is preferred when some neurons stop working / learning.



- The idea of Leaky ReLU can be extended further. Instead of

- The idea of Leaky ReLU can be extended further. Instead of multiplying z with a constant term, we can multiply it with a hyperparameter. This extension of leaky ReLU is known as Parametric ReLU.